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Studierichting: Informatica
Oefeningenreeks 1E nr. 1

Deel $z^5 + 3z^4 + 5z^3 - 9z^2 - 4z + 10$ door $z^2 + 1$

Stel:

$$z^5 + 3z^4 + 5z^3 - 9z^2 - 4z + 10 = (z^2 + 1)(az^3 + bz^2 + cz + d) + ez + f$$

$$(z^2 + 1)(az^3 + bz^2 + cz + d)$$

$$= az^5 + bz^4 + cz^3 + dz^2 + az^3 + bz^2 + cz + d$$

$$= az^5 + bz^4 + (c + a)z^3 + (d + b)z^2 + cz + d$$

$$= az^5 + bz^4 + (c + a)z^3 + (d + b)z^2 + (c + e)z + (d + f)$$

Vergelijk de coëfficiënten:

$$a = 1$$

$$b = 3$$

$$c + a = 5$$

$$d + b = -9$$

$$c + e = -4$$

$$d + f = 10$$

$$c + 1 = 5$$

$$c = 4$$

$$d + 3 = -9$$

$$d = -12$$

$$4 + e = -4$$

$$e = -8$$

$$-12 + f = 10$$

$$f = 22$$

Quotient: $z^3 + 3z^2 + 4z - 12$

Rest: $-8z + 22$

References